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SDITO Lab

Assignment No : 08

Problem Statement : Deploy a Project from Local Machine to GitHub and Vice-Versa.

Aim

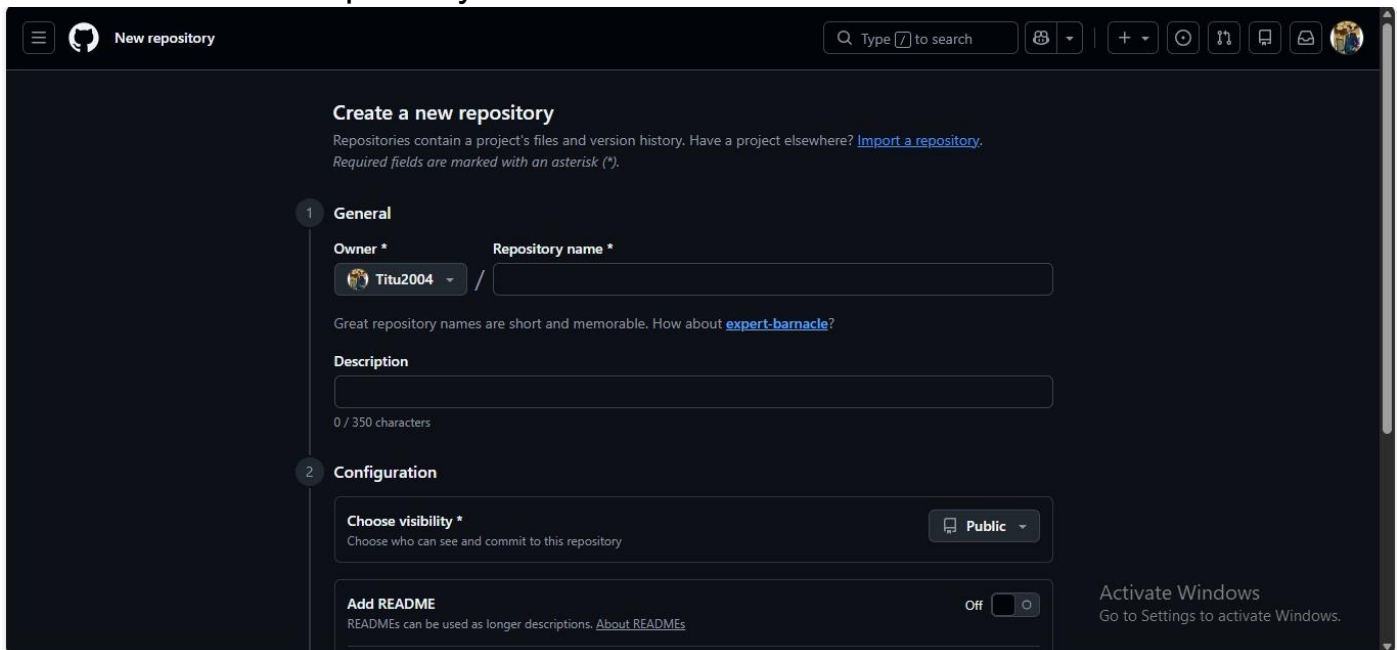
To upload a project from a local system to GitHub and then clone (download) it back from GitHub to the local machine using Git and GitHub authentication.

Part A: Deploy Project from Local Machine to GitHub.

Solution:

Step 1: Create a Repository on GitHub

- i. Login to GitHub.
- ii. Click on New Repository.
- iii. Enter repository name (example: **asg8**).
- iv. Set repository visibility to Public.
- v. Do NOT initialize with README.
- vi. Click Create repository



The screenshot shows the GitHub 'New repository' page. The page is titled 'Create a new repository' and includes a subtitle: 'Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository.](#) Required fields are marked with an asterisk (*).'

The page is divided into two main sections: 'General' and 'Configuration'.

General Section:

- Owner ***: A dropdown menu showing 'Titu2004'.
- Repository name ***: An empty text input field.
- Description**: An empty text input field with a character count '0 / 350 characters'.

Configuration Section:

- Choose visibility ***: A dropdown menu showing 'Public'.
- Add README**: A toggle switch set to 'Off'.
- Activate Windows**: A message 'Go to Settings to activate Windows.'

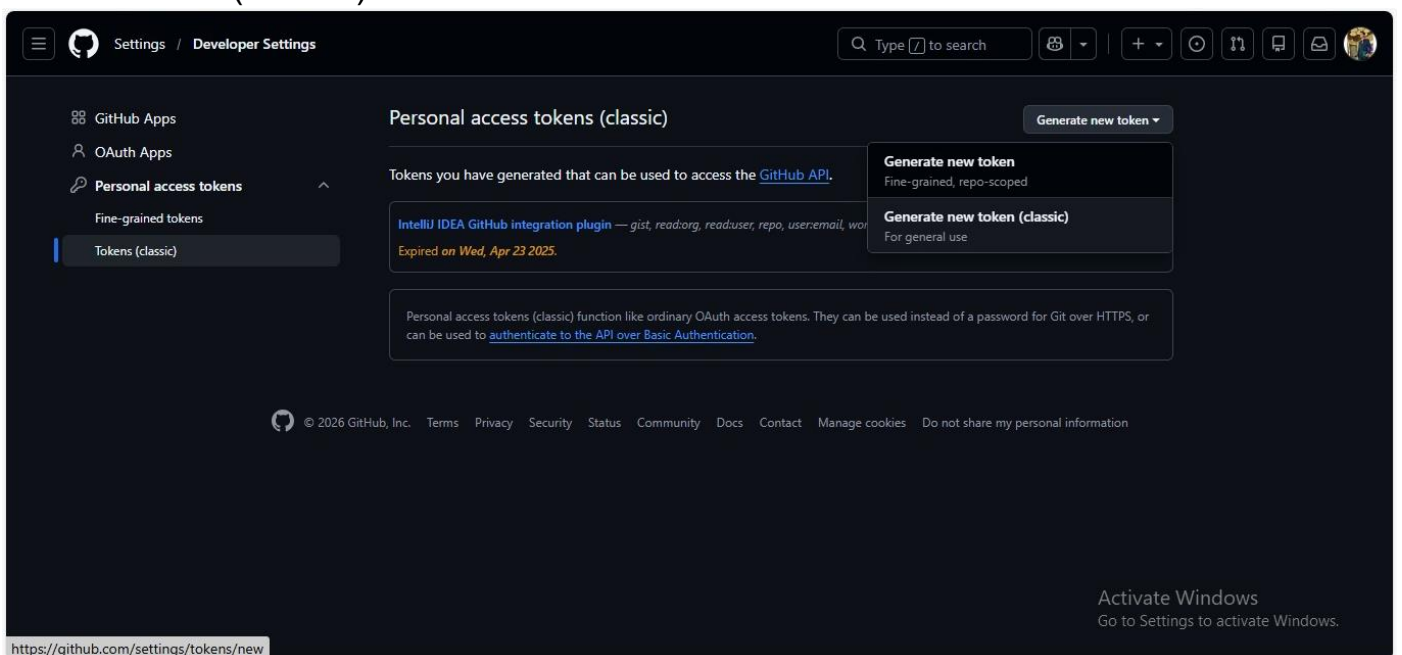
This screenshot shows the 'Configuration' step (step 2) of creating a new GitHub repository. It includes a 'Description' field with a 0/350 character count. Below this are three sections: 'Choose visibility' with a dropdown set to 'Public', 'Add README' with a toggle set to 'Off', and 'Add .gitignore' with a dropdown set to 'No .gitignore'. There is also an 'Add license' section with a 'No license' dropdown. A blue 'Create repository' button is at the bottom right. The footer shows the GitHub logo, copyright notice for 2026, and various links like Terms, Privacy, Security, Status, Community, Docs, Contact, and Manage cookies. An 'Activate Windows' watermark is visible in the bottom right corner.

Step 2: Generate GitHub Personal Access Token (PAT)

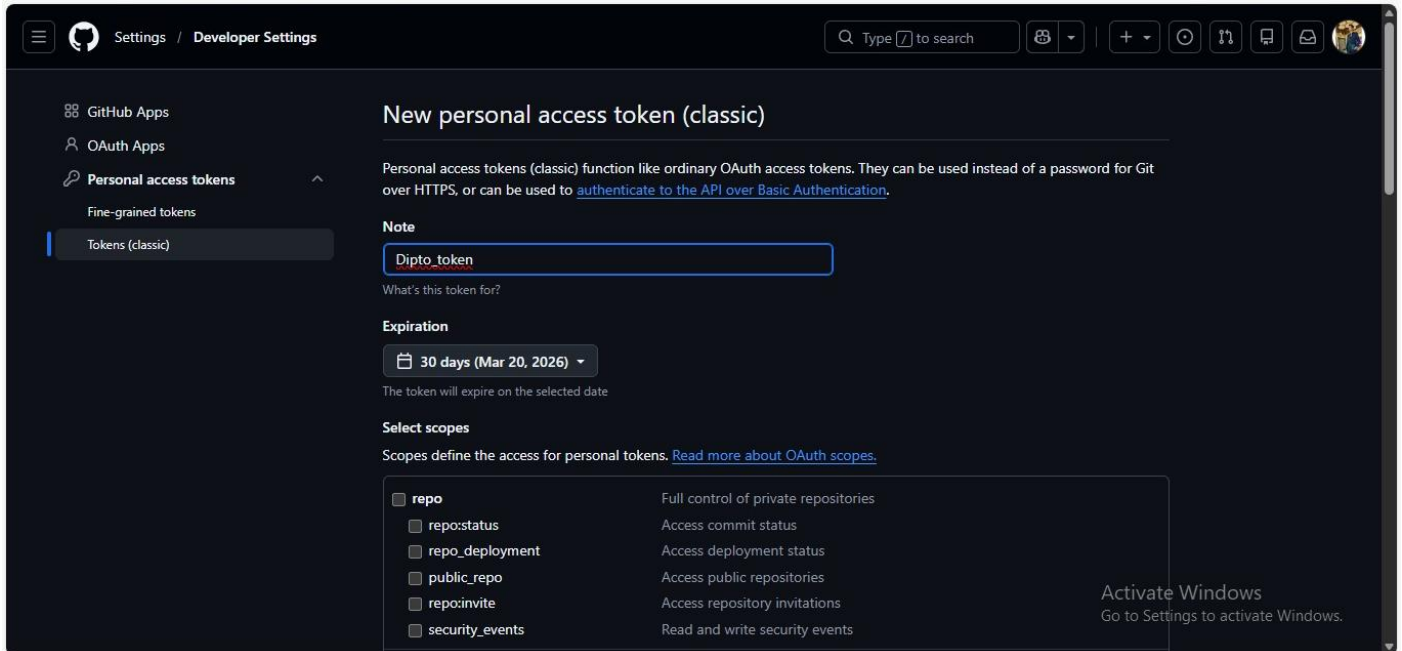
GitHub does **NOT** allow password authentication anymore.

Steps to create PAT

- i. Go to:
GitHub → Settings → Developer Settings → Personal access tokens → Tokens (classic)

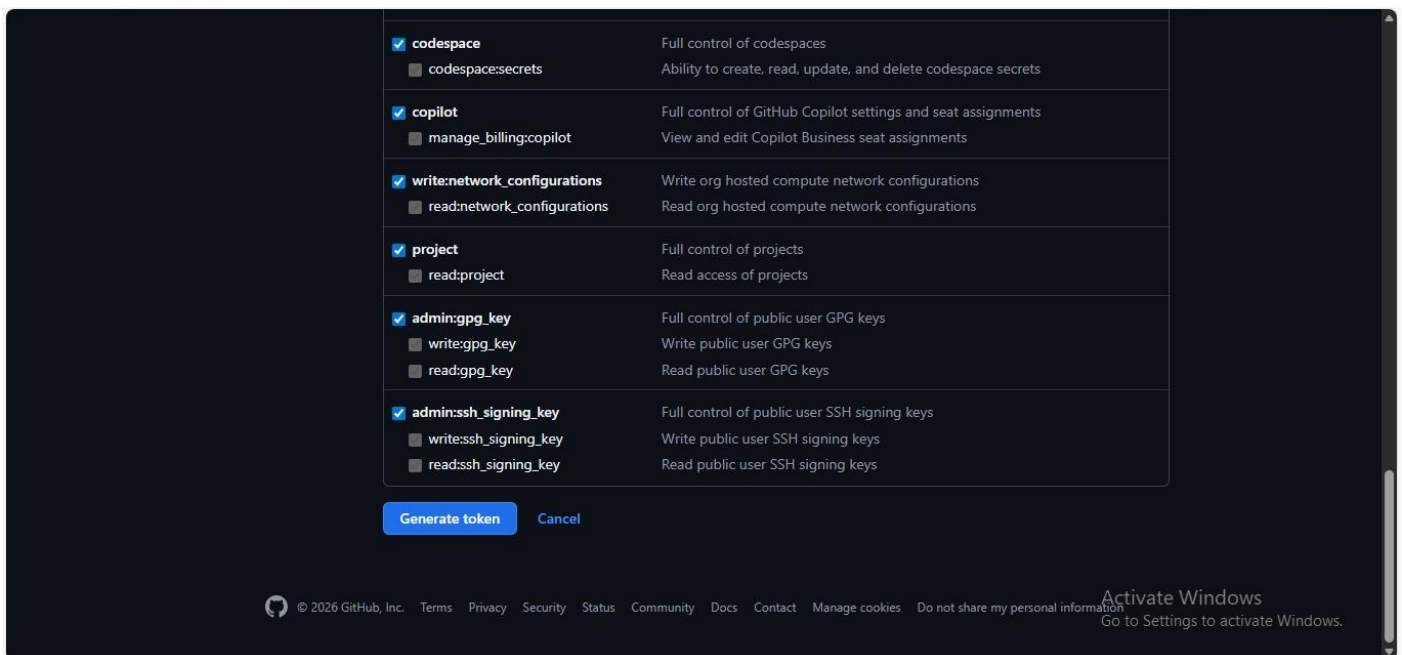


- ii. Click **Generate new token (classic)**
- iii. Fill:
 - **Note:** any name (e.g., **Dipto_token**)
 - **Expiration:** 90 days (or as required)



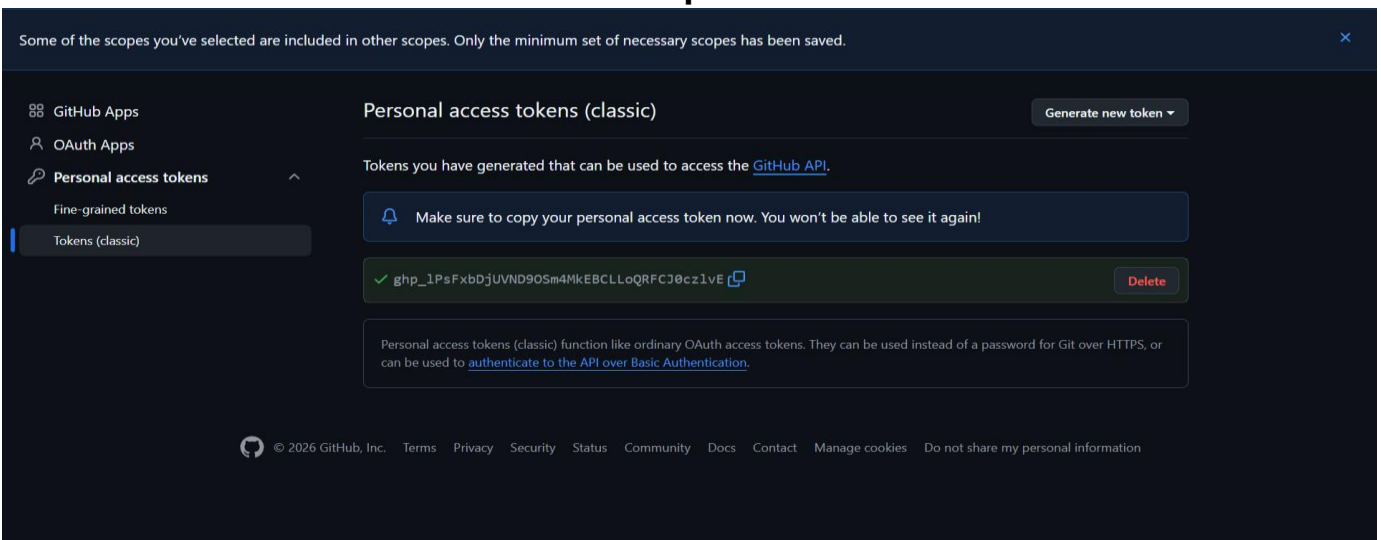
iv. Select all scopes.

v. Click **Generate token**



vi. **COPY THE TOKEN IMMEDIATELY**

This token will be used **instead of password**.



Step 2: Open Project Folder on Local Machine

- Open the folder (**webpage**) containing the project files.
- Right-click inside the folder and select Open in Terminal (or open Command Prompt / Git Bash and navigate to the folder).



```
MINGW64:/c/Users/User/OneDrive/Desktop/webpage
User@LAPTOP-SCS8GOAS MINGW64 ~/OneDrive/Desktop/webpage
$
```

Step 3: Initialize Git Repository

Run the command:

git init

This converts the local folder into a Git repository.

Step 4: Add Files to Staging Area

Add all project files:

git add .

Step 5: Commit the Files

Create the first commit:

git commit -m "Initial commit"

Step 6: Connect Local Repository to GitHub

Copy the repository URL from GitHub and run:

git remote add origin https://github.com/ Titu2004/asg8.git

Verify remote:

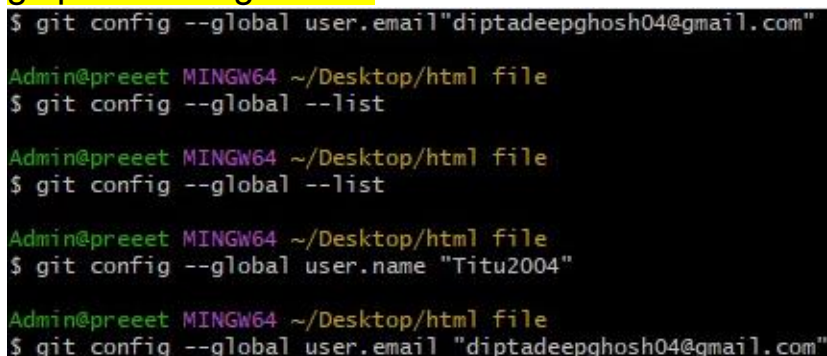
git remote -v

Step 7: Push Project to GitHub

Push the project to GitHub:

git branch -M main

git push -u origin main



```
$ git config --global user.email "diptadeepghosh04@gmail.com"
Admin@preet MINGW64 ~/Desktop/html file
$ git config --global --list
Admin@preet MINGW64 ~/Desktop/html file
$ git config --global --list
Admin@preet MINGW64 ~/Desktop/html file
$ git config --global user.name "Titu2004"
Admin@preet MINGW64 ~/Desktop/html file
$ git config --global user.email "diptadeepghosh04@gmail.com"
```

```

Admin@preeet MINGW64 ~/Desktop/html file
$ git config --global --list
user.name=Titu2004
user.email=diptadeepghosh04@gmail.com

Admin@preeet MINGW64 ~/Desktop/html file
$ git init
Initialized empty Git repository in C:/Users/Admin/Desktop/html file/.git/

Admin@preeet MINGW64 ~/Desktop/html file (master)
$ git add .

Admin@preeet MINGW64 ~/Desktop/html file (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   index.html
        new file:   next.html
        new file:   previous.html

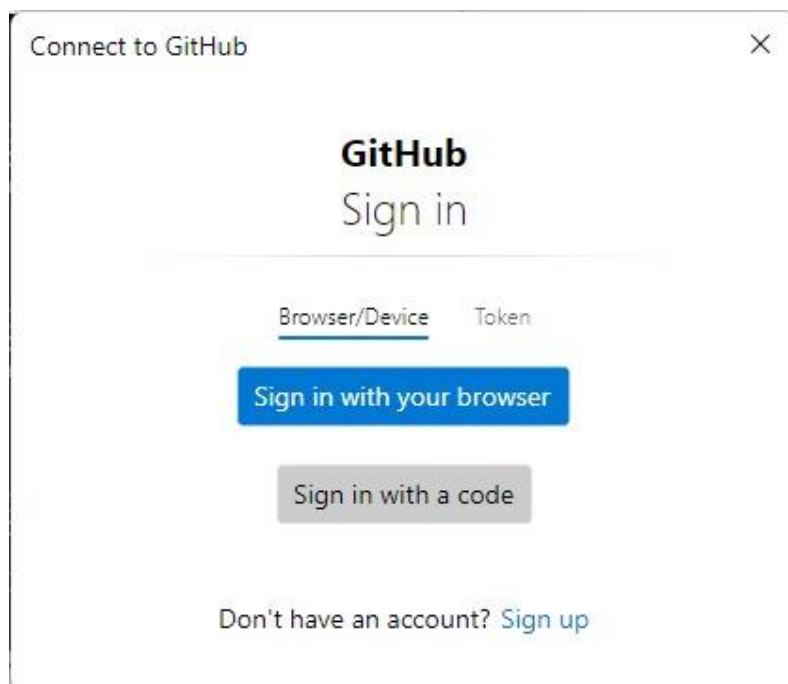
Admin@preeet MINGW64 ~/Desktop/html file (master)
$ git commit -m "done"
[master (root-commit) ae135df] done
3 files changed, 50 insertions(+)
create mode 100644 index.html
create mode 100644 next.html
create mode 100644 previous.html

Admin@preeet MINGW64 ~/Desktop/html file (master)
$ git remote add origin

```

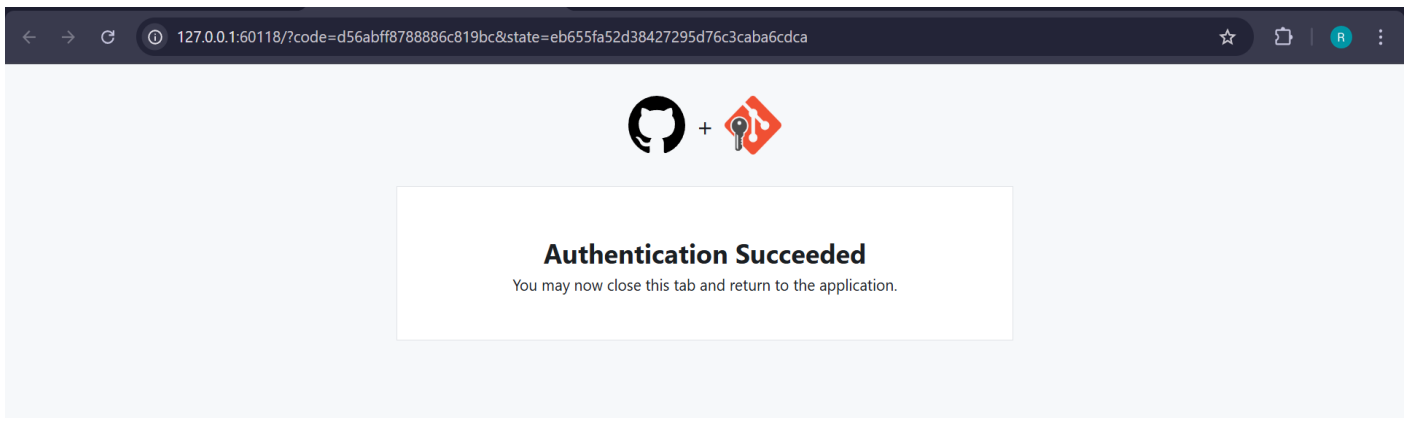
Step 8: Connectin to GitHub by Authentication

It will generate the sign in window of github by connecting from gitbash



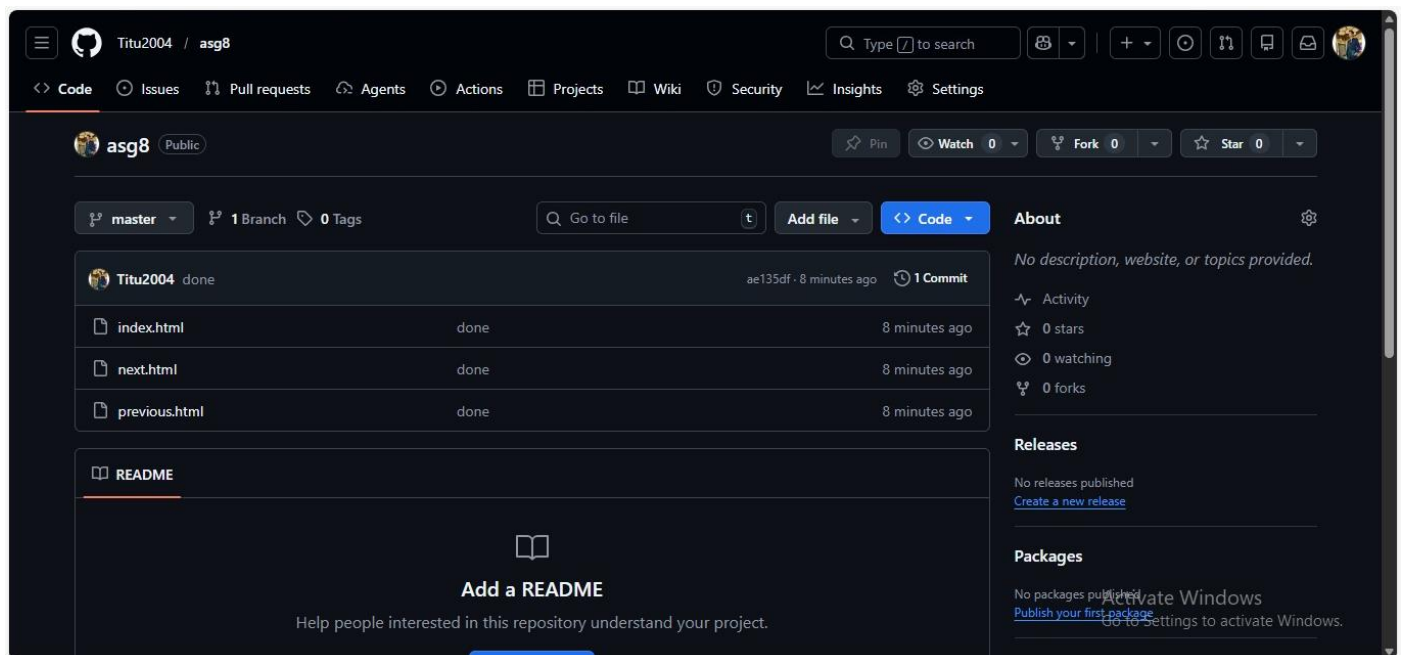
Step 9: Successefully connected

After sign in with sign in with browser or code or the copied token [personal access token (classic)], it will successfully connected with github and the project files of the folder(**webpage**) is deployed on the github successfully and by generating it will show "**Authentication Succeeded**" in new browser tab.



Step 10: Verify on GitHub

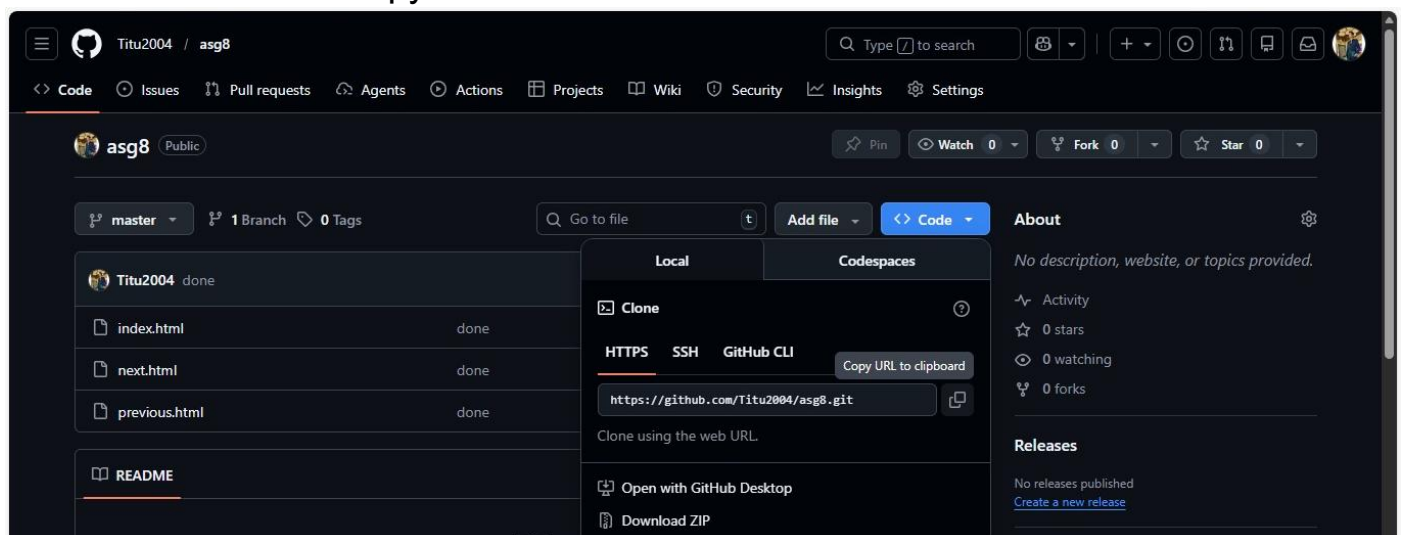
- Refresh repository page
- Files should be visible online



Part B: Deploy Project from GitHub to Local Machine

Step 11: Copy GitHub Repository URL

1. Open the GitHub repository.
2. Click Code → Copy HTTPS URL.



Step 12: Clone Repository to Local Machine

Open gitbash new local folder in desired directory and run:

```
git clone https://github.com/Titu2004/asg8.git
```

Step 13: Verify Project Files

Navigate into the cloned folder:

List files:

```
Admin@preeet MINGW64 ~/Desktop/New folder (2)
$ git clone https://github.com/Titu2004/asg8.git
Cloning into 'asg8'...
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (4/4), done.
remote: Total 5 (delta 1), reused 5 (delta 1), pack-reused 0 (from 0)
Receiving objects: 100% (5/5), done.
Resolving deltas: 100% (1/1), done.

Admin@preeet MINGW64 ~/Desktop/New folder (2)
$
```

Result

The project was successfully deployed from the local machine to GitHub and retrieved back from GitHub to the local machine using Git version control commands.